

FYP II

Final Year Project Reports

The culmination of your Final Year Project (FYP) II journey is marked by the submission of a comprehensive written report. This document, ranging from 90 to 150 pages in length, serves as a meticulously detailed account of your project's progression and achievements.

Following is summary of Report content and chapters:

- **Abstract**
- **List of Abbreviations**
- **List of Figures**
- **List of Tables**

- **Chapter 1 - Introduction**
 - Project Overview
 - Problem Description
 - Project Scope
 - Chapter wise Summary

Submission Deadline: 2nd week

- **Chapter 2 – Literature Review**

Submission Deadline: 3rd week

- **Chapter 3 – Requirement Specification**
 - Proposed System
 - Team Structure
 - Work Break Down Structure
 - Gantt Chart
 - Requirement Specifications
 - Functional Requirements
 - Non-Functional Requirements
 - Interface Requirements
 - Use Cases

- **Chapter 4 - System Design**
 - Design Considerations
 - Assumptions
 - Design Constraints
 - System Environment
 - Design Methodology
 - System Architecture
 - Overview (with diagram)

- Subsystem/Component/Module 1
 - Subsystem/Component/Module 2
 - Subsystem/Component/Module 3 ... Module N
- Conceptual Design
- Physical Design
- High Level Design
- Low Level Design
- Database Schema
 - Tables
 - Fields
 - Relationships
 - Entity Relationship Diagram (ERD)

Submission Deadline: 5th week

- **Chapter 5 – System Implementation**
 - Development Environment and Tools
 - Hardware & Software Specifications
 - Development Tools & Libraries
 - Implementation Details
 - Development Process
 - Coding Standards and Practices
 - Testing Strategies
 - Deployment and Configuration
 - Deployment Environment
 - Installation Procedures
- **Chapter 6 – System Testing & Evaluation**
 - Testing Strategy
 - Overview
 - Testing Levels
 - Testing Procedures
 - Test Cases
 - Test Execution
 - Test tools (if any)
 - Summary of Testing Effort
- **Chapter 7 – Conclusion**
 - Key Achievements
 - Project Significance and Impact
 - Contribution to the field
 - Impact on Future Work
 - Future Work
 - Summary of Learning Outcomes
 - Acknowledgement

- **References**

Submission Deadline: 8th week

Final Evaluation Guidelines for FYP Students

Evaluation Process

Your Final Year Project (FYP) will be evaluated by a designated external examiner, typically a professional from the IT industry. This evaluation ensures your project meets industry standards and addresses real-world applications. To ensure a successful evaluation, adhere to the following submission requirements and deadlines established by your department.

Submission Requirements

- **Poster/Standee:** Prepare a visually appealing poster or standee that effectively summarizes your project's essence. This will be displayed during the evaluation and should clearly communicate your project's goals, methodology, and key findings to a non-technical audience.
- **Project Book:** Submit a comprehensive project book that meticulously documents your project's journey. Refer to the department's guidelines for specific formatting and content requirements. Generally, this book should encompass a detailed literature review, your chosen methodology, the implementation process, achieved results, and the project's overall significance.
- **Plagiarism Report:** Ensure the originality of your work by submitting a plagiarism report generated by a reputable plagiarism detection tool. This demonstrates your commitment to academic integrity and ethical research practices.
- **Setup File (if applicable):** If your project requires a specific setup process, provide a well-documented setup file. This file should guide the evaluator through the installation and configuration steps necessary to execute your project.
- **Code (if applicable):** For projects involving software development, submit your well-commented and documented source code. Ensure the code adheres to established coding standards and best practices for readability and maintainability.
- **Running Software (if applicable):** Provide a functional version of your developed software. This could be a compiled executable, a web application accessible through a specific URL, or any other means that allows the evaluator to interact with your project's core functionalities.

FYP Report Styling and Formatting Guidelines

- **Chapter Headings:** Centered, font size 16.
- **Report Structure**
 - **Preface:** Included before the Abstract (with roman numeral page numbering).
 - **Table of Contents:** Follows the Preface (with roman numeral page numbering).
- **Font:** Times New Roman throughout.
- **Text Body**

- Font size 12, justified alignment.
- **Headers and Footers**
 - Header: Centered project title, font size 11 (on all pages).
 - Footer: Page numbers and footnotes (if required).
- **Headings**
 - Numbering: Follows chapter number (e.g., 1.1, 1.1.1, 1.1.2, etc.).
 - Font size: 14, no colons (:).
 - Subheadings: Font size 12.
- **Tables and Figures**
 - Numbering: Table 1, Table 2, etc. (Figure 1, Figure 2, etc.).
 - Placement
 - Table titles and numbers: Above the table.
 - Figure captions: Centered, below the figure.
 - Font size
 - Table titles: 12.
 - Figure captions: 10.
- **Citations and References**
 - Use in-text citations.
 - Reference list: Saved for insertion in the final "References" chapter.

FYP Report Content Explanation

Abstract: The abstract serves as a concise yet informative overview of your entire Final Year Project (FYP) report. It acts as a first impression for readers, enticing them to delve deeper into the details of your project.

- **Structure:** An effective abstract typically follows a three-part structure:
 - **Introduction:** Briefly introduce project topic, highlighting the problem addressed or the opportunity explored.
 - **Body:** Summarize the key aspects of project, including:
 - Research objectives or project goals.
 - The methodology group has employed (e.g., the techniques and processes used).
 - The main findings or results achieved by the project group.
 - **Conclusion:** Conclude by emphasizing the significance of project's findings and their potential impact on the IT field.
- **Length:** An abstract typically ranges from 150-250 words.

List of Abbreviations

- **Purpose:** If you utilize a significant number of abbreviations throughout your report, a dedicated "List of Abbreviations" section is crucial for reader comprehension. This section ensures clarity and avoids confusion for those unfamiliar with the specific acronyms or abbreviations used.

- **Content:** This list should alphabetically order the abbreviations employed within the report. For each abbreviation, project group must provide its corresponding full definition or explanation.
- **Placement:** The "List of Abbreviations" section is typically placed after the Abstract and before the Table of Contents.

Guideline for Maintaining List of Abbreviations



LIST OF ABBREVIATIONS	
CIDSA	Coremfengiatcras idligulacestibulum doloripurusmetus spsuntortor armctnuncad
CIN	Cestibulum in nunc
CMT	Cras metus tortor
GNQ	Givamus non quam
IA	Isuspendisse aliquam
IFLID	Iroindapibu feugiateleifend liguladipurasadipiscing igloremipsum ddolorsiametconsectetur
LR	Led rhoncus
MCPJ	Muis cursus posuere justo

1. Include the heading "LIST OF ABBREVIATIONS" in all capital letters, and center it 1" below the top of the page.
2. Include one double-spaced line between the heading and the first entry.
3. Arrange your abbreviations alphabetically.
4. Abbreviations must align with the document's left margin or be indented to the right of the left page margin using consistent tabs.
5. If an entry takes up more than one line, single-space between the two lines.
6. Include one double-spaced line between each entry.
7. The List of Abbreviations page(s) must be numbered with consecutive lower case Roman numerals centered with a 1/2" margin from the bottom edge.

Tips

- **Clarity and Concision:** Strive for clear and concise language throughout your abstract and list of abbreviations.
- **Keywords:** Consider incorporating relevant keywords within your abstract to enhance searchability of your project report.

List of Figures and List of Tables

Both the List of Figures and List of Tables serve as essential navigational tools within Final Year Project (FYP) report. They act as a quick reference guide, enabling readers to locate specific figures and tables of interest effortlessly.

List of Figures: This section meticulously enumerates all figures included within your report. Figures encompass visual elements such as charts, graphs, diagrams, or illustrations.

- **Content:** Each entry in the List of Figures should include the following:
 - **Figure Number:** Assign a unique sequential number to each figure (e.g., Figure 1, Figure 2, etc.).
 - **Figure Caption:** Replicate the exact caption you used for the corresponding figure within the report itself. Captions typically provide a concise and informative description of the figure's content.
 - **Page Number:** Indicate the specific page where the corresponding figure is located within the report.

List of Tables: Similar to the List of Figures, this section serves as an index for all tables included in your report. Tables present data in a structured and organized format, often employing rows and columns.

- **Content:** Each entry in the List of Tables should include the following:
 - **Table Number:** Assign a unique sequential number to each table (e.g., Table 1, Table 2, etc.).
 - **Table Title:** Replicate the exact title you used for the corresponding table within the report itself. Titles should accurately reflect the content presented within the table.
 - **Page Number:** Indicate the specific page where the corresponding table is located within the report.

Placement: Both the List of Figures and List of Tables are typically positioned after the Table of Contents and before the first chapter of your report.

Tips

- **Consistency:** Ensure consistency between the numbering and captions used in the Lists and the actual figures and tables within the report's body.
- **Organization:** Maintain alphabetical order for figure captions or a logical order based on appearance within the report for table titles within the respective lists.

Details of Chapters in Report

Chapter 1 - Introduction

Here's a breakdown of the recommended structure for Chapter 1: Introduction:

1.1 Introduction (8-10 lines): This introductory paragraph serves as a roadmap for the chapter, outlining the key topics that will be discussed in detail throughout Chapter 1. Briefly introduce your project, highlighting the problem you aim to address and the proposed solution.

1.2 Project Overview: The Project Overview section provides a comprehensive picture of the project, setting the context for the reader. This section should encompass:

- A concise overview of the project's purpose and goals.
- The specific problem you are addressing within the IT field.
- The potential impact of your project and its anticipated contribution to the field.

1.2.1 Problem Description: (Subheading within Project Overview) This section delves deeper into the specific problem that project tackles. Here, group should meticulously describe:

- The nature of the IT challenge you aim to address.
- The significance of this problem within the broader IT landscape.
- Any existing solutions or approaches, highlighting their limitations.

1.3 Project Scope: The Project Scope section defines the boundaries of your project. This section should clearly outline:

- The specific deliverables group aims to achieve.
- The functionalities and features the project will encompass.
- Any limitations or exclusions within the project's scope.

Chapter-wise Summary: Explicitly mention in this structure, including a concise chapter summary at the end of this chapter. This can reiterate the key points covered and serve as a quick reference point for readers.

Chapter 2: Literature Review

Each team of students should submit atleast 20 pages of chapter 2. The content must be original, not copied, so rephrase the content and cite the source from where you have taken it.

A literature review is a survey of scholarly sources on a specific topic. It provides an overview of current knowledge, allowing you to identify relevant theories, methods, and gaps in the existing research. The Literature Review chapter paints a comprehensive picture of the existing landscape within your chosen IT field. It demonstrates your knowledge, your critical thinking skills, and ultimately justifies the significance and originality of your project.

Writing a literature review involves finding relevant publications (such as books and journal articles), critically analyzing them, and explaining what you found. There are five key steps:

1. **Search** for relevant literature
2. **Evaluate** sources
3. **Identify** themes, debates and gaps
4. **Outline** the structure
5. **Write** your literature review

A good literature review doesn't just summarize sources – it analyzes, synthesizes, and critically evaluates to give a clear picture of the state of knowledge on the subject.

Following is the structure of Literature review:

2.1 Literature Review

After chapter heading write few lines (about 4-6 lines) about the description of the chapter i.e. what is this chapter all about.

Categorize the sources that you would be covering in literature review:

- Books
- Articles
- Applications/software
- Websites
- Research Papers

(You can add any other source as well if you are using in your report)

Above would be the main headings. Under these main headings use sub-headings to represent the title of the source and its screenshot:

2.1 Books

(2-3 lines about this section)

2.1.1 Book Name 1

Screenshot of the title of the book

Summary of the book

2.1.2 Book Name 2

Screenshot of the title of the book

Summary of the book

2.2 Articles

(2-3 lines about this section)

2.2.1 Article Title 1

Screenshot of the article title if available

Summary of the article

Similarly for other sources

Conclusion: (8-10 lines of the conclusion of this chapter.)

Chapter 3 – Requirement Specification

The Requirements Specification chapter blueprints your project. It meticulously outlines the functionalities, features, and performance expectations your project must deliver. In essence, it defines the "what" and "how" of your project, ensuring all stakeholders are aligned on the desired outcome.

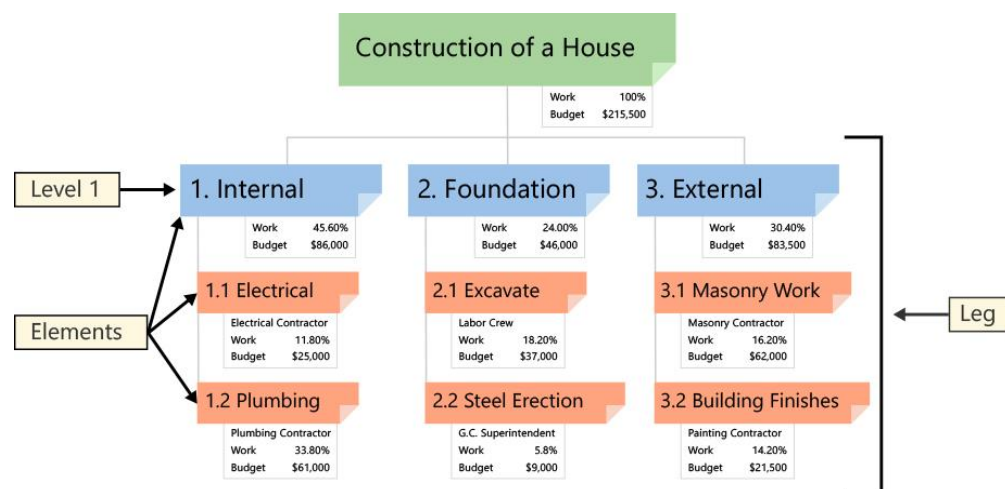
Each team of students should submit atleast 30 pages of chapter 3. The content must be original, not copied, so rephrase the content and cite the source from where you have taken it.

3.1 Proposed System: Proposed system means explaining what and how you are going to solve the identified real-world problem in your project. What is your project all about and what is new in your project other than existing things.

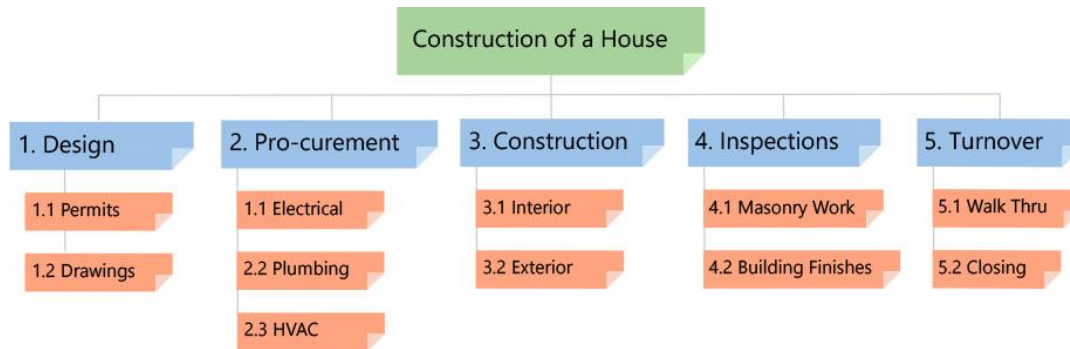
3.2 Team Structure: You need to describe what team is and its importance in a project. Next important thing is defining and describing the structure of your team for example you are following chief programmer team structure or democratic team structure. Explain the type of structure that you are following in your project and describe each team member's role in project.

3.3 Work Break Down Structure: First describe what is WBS and then the two types of it: Project-based WBS and Product-based WBS. Breaking work into smaller tasks is a common productivity technique used to make the work more manageable and approachable. For projects, the **Work Breakdown Structure (WBS)** is the tool that utilizes this technique and is one of the most important project management documents. It singlehandedly integrates scope, cost and schedule baselines ensuring that project plans are in alignment.

A Deliverable-Based Work Breakdown Structure or Product-based WBS clearly demonstrates the relationship between the project deliverables (i.e., products, services or results) and the scope (i.e., work to be executed). Following figure is an example of a Deliverable-Based WBS for building a house.



In following figure, a Phase-Based WBS or Project-based WBS, the Level 1 has five Elements. Each of these Elements are typical phases of a project. The Level 2 Elements are the unique deliverables in each phase. Regardless of the type of WBS, the lower Level Elements are all deliverables. Notice that Elements in different Legs have the same name. A Phase-Based WBS requires work associated with multiple elements be divided into the work unique to each Level 1 Element. A WBS Dictionary is created to describe the work in each Element.



3.4 Gantt Chart: A Gantt chart is a type of bar chart that illustrates a project schedule, named after its inventor, Henry Gantt, who designed such a chart around the years 1910–1915. Modern Gantt charts also show the dependency relationships between activities and current schedule status.

3.5 Requirement Specifications: A software requirements specification (SRS) is a document that describes what the software will do and how it will be expected to perform. It also describes the functionality the product needs to fulfill all stakeholders (business, users) needs. You need to mention:

3.5.1 Purpose. Definitions, system overview, and background.

3.5.2 Overall description. Assumptions, constraints, business rules, and product vision.

3.5.3 System attributes

3.5.4 Database requirements

3.5.5 Functional Requirements: A **Functional Requirement (FR)** is a description of the service that the software must offer. It describes a software system or its component. A function is nothing but inputs to the software system, its behavior, and outputs. It can be a calculation, data manipulation, business process, user interaction, or any other specific functionality which defines what function a system is likely to perform. Functional Requirements are also called **Functional Specification**.

Functional Requirement No.	Function Requirement Description
FR 1	User should be able to enter Sales Data
FR 2	Sales Reports should be generated every 24 hours
FR 3	API interface to Invoice System

Functional requirements must include following:

- Details of operations conducted in every screen
- Data handling logic should be entered into the system
- It should have descriptions of system reports or other outputs
- Complete information about the workflows performed by the system
- It should clearly define who will be allowed to create/modify/delete the data in the system
- How the system will fulfill applicable regulatory and compliance needs should be captured in the functional document

3.5.6 Non-Functional Requirements: Describe the general characteristics of a system. They are also known as quality attributes.

3.5.6.1 Performance and scalability. How fast does the system return results? How much will this performance change with higher workloads?

3.5.6.2 Portability and compatibility. Which hardware, operating systems, browsers, and their versions does the software run on? Does it conflict with other applications and processes within these environments?

3.5.6.3 Reliability, availability, maintainability. How often does the system experience critical failures? and how much time is it available to users against downtimes?

3.5.6.4 Security. How are the system and its data protected against attacks?

3.5.6.5 Localization. Does the system match local specifics?

3.5.6.6 Usability. How easy is it for a customer to use the system?

3.5.7 Interface Requirements: The Interface Requirements section within your Requirements Specification chapter focuses on how users will interact with your project. It details the user interface (UI) elements, ensuring a smooth and intuitive user experience. Think of it as the blueprint for how users will see and interact with your system.

3.6 Use Cases: The Use Cases section in your Requirements Specification chapter depicts real-world scenarios of how users will interact with your project to achieve specific goals. It essentially describes the "who," "what," "when," and "why" of user interactions, ensuring your project caters to a variety of user needs and functionalities. Must have use-case diagrams and descriptions of diagrams.

Chapter 4 - System Design

The System Design chapter serves as the architectural blueprint of your project. It depicts the high-level structure and organization of your system, detailing how various components will interact to achieve the desired functionalities. In essence, it translates your requirements into a concrete technical design that guides the development process.

4.1 Design Considerations: There are many aspects to consider in the design of a piece of software. The importance of each consideration should reflect the goals and expectations that the software is being created to meet. Some of these aspects are: This allows division of work in a software development project.

4.1.1 Assumptions: An assumption is defined as “a thing that is accepted as true or as certain to happen, without proof” or as “a fact or statement taken for granted”. Accordingly, we define software assumption as software development knowledge taken for granted or accepted as true without evidence.

4.1.2 Constraints: The *constraints* may be imposed on the hardware, *software*, data, operational procedures, interfaces, or any other part of the system. *Examples* may include a *constraint* that the system must use predefined COTS hardware or *software*, use of a particular algorithm, or implement a specific interface protocol. The definition of a constraint is something that imposes a limit or restriction or that prevents something from occurring. An example of a constraint is the fact that there are only so many hours in a day to accomplish things.

4.1.3 System Environment: where the application would run or the configurations and supporting tools and softwares required to run the application.

4.1.4 Design Methodology: Summarize the approach that will be used to create and evolve the designs for this system. Cover any processes, conventions, policies, techniques or other issues which will guide design work.

4.2 System Architecture: The architecture provides the top-level design view of a system and provides a basis for more detailed design work. Provide or reference a detailed description and diagrams of the architecture.

4.2.1 Overview: This section provides a high-level overview of the structural and functional decomposition of the system. Focus on how and why the system was decomposed in a particular way rather than on details of the particular components. Include information on the major responsibilities and roles the system (or portions) must play.

4.2.1.1 Subsystem/Component/Module 1: its description and if it is subdivided into further sub-modules.

4.2.1.2 Subsystem/Component/Module 2: its description and if it is subdivided into further sub-modules.

4.2.1.3 Subsystem/Component/Module N: its description and if it is subdivided into further sub-modules.

4.3 Conceptual Design: Conceptual design of a database means the early stage of a design process which includes design interactions, experience and strategies. Conceptual

Design consists of more than just describing small fragments of the application in terms of ER [Entity Relationship] diagrams.

4.4 Physical Design: Physical design relates to the actual input and output processes of the system. It focuses on how data is entered into a system, verified, processed, and displayed as output. It produces the working system by defining the design specification that specifies exactly what the candidate system does.

4.5 High Level Design: High-level design explains the architecture that would be used for developing a software product. The architecture diagram provides an overview of an entire system, identifying the main components that would be developed for the product and their interfaces.

4.6 Low Level Design: Low-level design is a component-level design process that follows a step-by-step refinement process.

4.7 Database Schema

The database schema section within System Design chapter outlines the structure of project's database. It defines the tables, columns, and relationships between them, essentially acting as a blueprint for how data will be stored and organized within your system. This ensures efficient data management and retrieval.

4.7.1 Tables: Provide a description of tables.

4.7.2 Fields: Provide a description of fields.

Table 1: List of Tables and Fields

Table Name	Field Name	Data Type	Allow Nulls	Field Description
BinderRequest	SellingRep	Varchar(50)		Get this field from BankPlan.ProposalRepCode But use actual name (Roy Pinnell)
BinderRequest	SigningRep	Varchar(50)		Get this field from Bankplan.SigningRepCode But use actual name (Roy Pinnell)
ProcessBP	ParentProcessID	Int	Yes	This will tie a subprocess to a process
BinderRequest	ProjectedWireDate	Date		The earliest Policy.ProposalWireDate of all included policies in the scenario.

4.7.3 Relationships: Provide a description of relationships that need to be created for the design.

4.7.4 Entity Relationship Diagram

4.8 UML Diagrams (see the UML book for different UMLs that you are required to draw for your system including class diagrams, use-case diagrams etc.)

Chapter 5: System Implementation

The System Implementation chapter of Final Year Project (FYP) report serves as a critical section, detailing the process of transforming conceptual project design into a functioning reality. Here's a breakdown of the key elements that should be included:

5.1 Development Environment and Tools

5.1.1 Hardware and Software Specifications: Outline the specific hardware and software resources utilized for project development. This could encompass details such as operating systems, programming languages, development platforms, and any specialized hardware requirements.

5.1.2 Development Tools and Libraries: List the various software development tools and libraries that facilitated the development process. This might include Integrated Development Environments (IDEs), version control systems, testing frameworks, or any specialized libraries relevant to your project's domain.

5.2 Implementation Details

5.2.1 Development Process: Offer a detailed account of the development process, outlining the employed methodologies (e.g., Agile, Waterfall) and any project management techniques used to ensure efficient development.

5.2.2 Coding Standards and Practices: Describe the coding standards and practices adhered to throughout the development process. This emphasizes your commitment to code quality, readability, and maintainability.

5.2.3 Testing Strategies: Outline the implemented testing strategies employed to ensure the system's functionality, performance, and security. This could include unit testing, integration testing, and system testing methodologies.

5.3 Deployment and Configuration

5.3.1 Deployment Environment: Describe the target environment for system deployment. This might encompass a local machine, a cloud server, or a network infrastructure.

5.3.2 Installation Procedures: Detail the steps involved in installing and configuring the system on the target environment. This section should be written with potential future users in mind, ensuring a smooth deployment process.

Chapter 6: System Testing and Evaluation

The System Testing and Evaluation chapter Final Year Project (FYP) report serves a crucial purpose. Here, the effectiveness of developed system is demonstrated by outlining the thorough testing procedures it underwent.

6.1 Testing Strategy

6.1.1 Overview: Begin by providing a concise overview of the overall testing strategy employed. Explain the testing methodologies chosen (e.g., black-box testing, white-box testing), and how these methodologies align with your project's specific needs.

6.1.2 Testing Levels: Delineate the various testing levels implemented, such as unit testing, integration testing, and system testing. Describe the purpose and scope of each testing level.

6.2 Testing Procedures

6.2.1 Test Cases: Outline the development of detailed test cases for each system module and functionality. Test cases should encompass both positive and negative scenarios to comprehensively evaluate system behavior.

6.2.2 Test Execution: Describe the process of test execution. This could involve manual testing, automated testing frameworks, or a combination of both. Explain how test results were documented and tracked.

6.2.3 Test Tool (if any)

6.3 Summary of Testing Efforts: Summarize the effectiveness of your testing strategy. Reiterate how testing contributed to the development of a robust and reliable system.

Additional Considerations

- Include relevant tables, figures, and screenshots to visually represent test results and system performance metrics. Ensure these visuals are clear, concise, and easy to understand.
- Maintain a professional tone throughout the chapter, focusing on factual reporting and avoiding overly technical jargon.

Chapter 7: Conclusion

The Conclusion chapter of the report serves as a pivotal opportunity to summarize project's journey, reiterate its significance, and leave a lasting impression on the reader. Here's a breakdown of the key elements that can be incorporated:

7.1 Key Achievements: Summarize the project's key achievements, emphasizing the successful implementation of your proposed solution. Quantifiable results or metrics can be particularly impactful here.

7.2 Project Significance and Impact

7.2.1 Contribution to the Field: Explain how the project contributes to the advancement of the chosen IT field. Discuss the potential real-world applications of the developed system or the theoretical insights gleaned from research.

7.2.2 Impact on Future Work: Highlight potential avenues for future research or development based on project's findings. This demonstrates the project's potential to inspire further exploration and advancement within the field.

7.3 Future Work: Suggest potential improvements or future research directions.

7.4 Summary of Learning Outcomes: Share some of the key learnings and insights you gained throughout the FYP experience. This personal reflection demonstrates your intellectual growth and the value derived from the project.

7.5 Acknowledgement: Express your gratitude to your supervisor, mentors, or any individuals who provided guidance and support during the project's development.

REFERENCES